39 Giving a Voice to the Altamura Man: Artificial Intelligence in Museum Engagement

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The application of Artificial Intelligence to museum communication presents a wide range of uses and potential. Promoted by the Aldo Moro University of Bari in collaboration with the National Archaeological Museum of Altamura, and funded by DICOLAB – Cultura al Digitale through a research fellowship held by the author, the research project entitled Giving Voice to the Altamura Man – Artificial Intelligence in Museum Engagement aimed to study and implement these potential applications, allowing visitors to literally communicate with the museum exhibition. Conceived by the Director of the Museum, Dr. Ebe Chiara Principalli, the project focused on the creation of an AI connected to a live avatar, designed to give voice to the hyper-realistic mannequin of the Altamura Man on display at the museum, thereby enabling a direct dialogue with visitors. From its earliest stages, the development of this AI required an interdisciplinary approach. Under the scientific supervision of Prof. Giuliano de Felice, Prof. Giulio Palumbi, and Dr. Ebe Chiara Principalli, the archaeological contribution involved gathering existing literature on Neanderthals and selecting specific texts to ensure the AI's language remained scientifically accurate. Simultaneously, technical training in the creation and application of AI to cultural heritage was coordinated by Prof. Gino Roncaglia. These combined competencies converged into the definition of the AI's intent, assigning it a specific role and persona: a 9-year-old Neanderthal girl named Lìa. The procedural methodology—structured to identify the most suitable tools for creating the live avatar, resolve challenges related to interdisciplinary collaboration, and systematize texts for AI training—provides a valuable methodological guideline. This framework is valuable both for application to other contexts of public engagement with archaeological heritage and for the finalization of the project, which is currently nearing completion.